

# Database Management Systems

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# RAID

# RAID

- The choice of **disk structure** is very important in databases. Important factors, besides price, are:
  - **Capacity**
  - **Speed**
  - **Reliability**
- **RAID: Redundant Arrays of Independent Disks**
  - disk organization techniques that manage a large numbers of disks, providing a view of a single disk of
    - ▶ **high capacity** and **high speed** by using multiple disks in parallel, and
    - ▶ **high reliability** by storing data redundantly, so that data can be recovered even if a disk fails

# Improvement of Reliability via Redundancy

- **Redundancy** – store extra information that can be used to rebuild information lost in a disk failure
- E.g., **Mirroring** (or **shadowing**) – **RAID level 1**
  - Duplicate every disk. Logical disk consists of two physical disks.
  - Every write is carried out on both disks
    - ▶ Reads can take place from either disk
  - If one disk in a pair fails, data still available in the other
    - ▶ Data loss would occur only if a disk fails, and its mirror disk also fails before the system is repaired
      - Probability of combined event is very small, (except for dependent failure modes such as fire or building collapse or electrical power surges)
- **Mean time to data loss** depends on mean time to failure, and **mean time to repair**

# Mean Time to Failure (MTTF)

## Definition

MTTF is the **average time a disk or system operates before it fails.**

It is usually measured in:

- Hours
- Days
- Years

For hard disks, manufacturers often specify values like:

- 1,000,000 hours
- 1.5 million hours
- 2 million hours

## Example

Suppose:

- One HDD has an MTTF of **1,000,000 hours**

This means:

On average, the disk is expected to fail once every 1,000,000 operating hours.

It does **not** mean the disk will definitely last 1,000,000 hours.

## RAID Example

Assume:

- RAID 1 contains **2 disks**
  - Each disk MTTF = 1,000,000 hours
- Since either disk can fail, the RAID experiences a disk failure more frequently.

Approximate RAID failure rate:

$$\begin{aligned}\text{RAID MTTF} &= \frac{1,000,000}{2} \\ &= 500,000 \text{ hours}\end{aligned}$$

- In RAID with more disks,
- chance of one disk failing increases.

# Improvement in Performance via Parallelism

- Two main goals of parallelism in a disk system:
  1. Load balance multiple **small accesses** to increase throughput
  2. Parallelize large accesses to reduce response time.
- Improve transfer rate by **striping data** across multiple disks.

```
User A ┘  
User B ┣→ Disk 1 (busy handling all requests)  
User C ┘
```

## Multiple Disk System

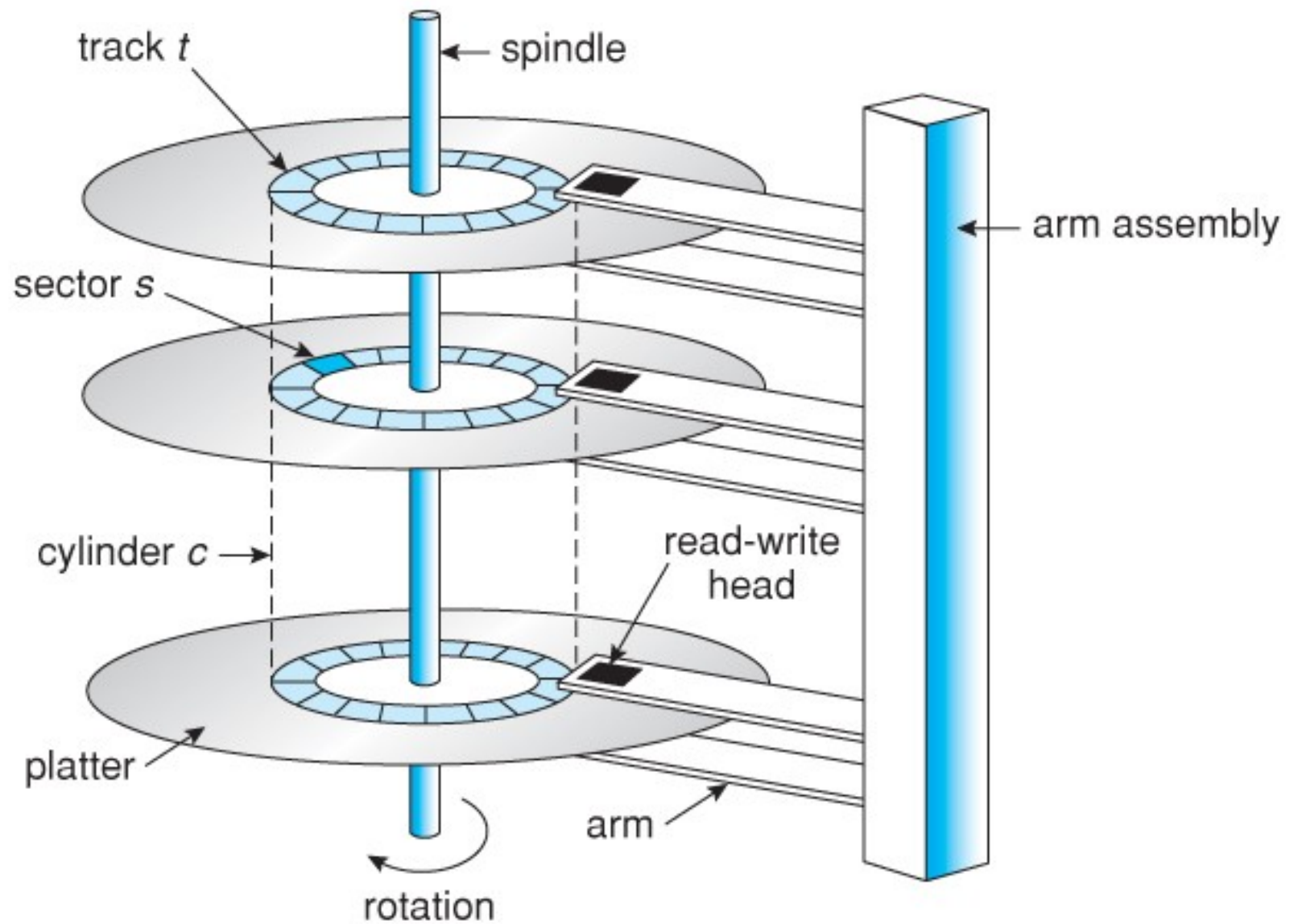
```
User A → Disk 1  
User B → Disk 2  
User C → Disk 3
```

# Improvement in Performance via Parallelism

- Improve transfer rate by **striping data** across multiple disks.
- **Bit-level striping** – split the bits of each byte across multiple disks
  - In an array of eight disks, write bit *i* of each byte to disk *i*.
  - Each access can read **data at eight times the rate of a single disk**.
  - But seek/access time worse than for a single disk
    - ▶ Bit level striping is not used much any more

# Improvement in Performance via Parallelism

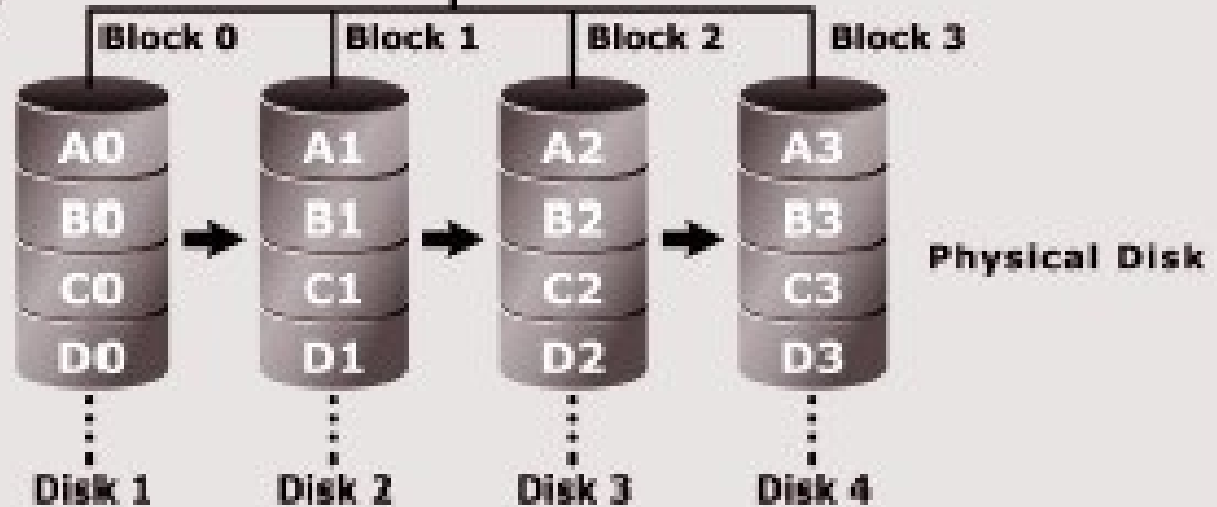
- **Block-level striping** – with  $n$  disks, block  $i$  of a file goes to disk  $(i \bmod n) + 1$ 
  - Requests for different blocks can run in parallel if the blocks reside on different disks
  - A request for **a long sequence of blocks can utilize all disks in parallel**
  - Usually called **RAID level 0**





**Array  
Management**

**RAID 0 : Striping**

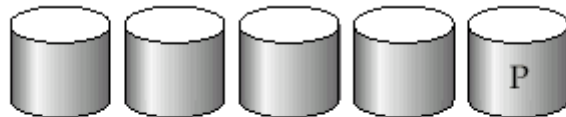


# RAID Levels 2 and 3

- **RAID Level 2: Memory-Style Error-Correcting-Codes (ECC)** with bit striping.
- **RAID Level 3: Bit-Interleaved Parity**
  - a single parity bit is enough for error correction, not just detection, since we know which disk has failed
    - ▶ When writing data, corresponding parity bits must also be computed and written to a parity bit disk
    - ▶ To recover data in a damaged disk, compute XOR of bits from other disks (including parity bit disk)



(c) RAID 2: memory-style error-correcting codes



(d) RAID 3: bit-interleaved parity

Suppose:

- Data bits =  $m$
- ECC bits =  $r$

Then:

$$2^r \geq m + r + 1$$

# Choice of RAID Level

- Mirroring provides much better write performance than Parity
  - For Parity read operations are needed before a write, whereas Mirroring only requires 2 writes
  - Mirroring preferred for high update environments such as log disks
- Mirroring needs more disks for the same capacity
  - Higher cost per capacity unit
    - ▶ But price of storage capacity is already low and decreasing
- Nowadays, for database systems
  - (Distributed) Parity, with RAID Level 5, is preferred for applications with low update rate, and large amounts of data
  - Mirroring is preferred for all other applications
- The choice of the storage structure is an important first step when designing a real database!

# RAID Implementations

- ▶ Hardware (usually a specialized disk controller card)
  - Controls all drives attached to it
  - Array(s) appear to host operating system as a regular disk drive
  - Provided with administrative software
- ▶ Software
  - Runs as part of the operating system
  - Performance is dependent on CPU workload
  - Does not support all RAID levels

